MAWHIBA GRADE 9 · UNIT 3 LINEAR FUNCTIONS AND RELATIONS · ACTIVITY 2

Paired Variables

Advanced Supplementary Mathematics
Mawhiba Schools Partnership

Student ..... **Class** ..... **Date** ..... / ..... / 2026 **Teacher** Mr Malek Thiab

UNIT 3 LEARNING OBJECTIVES

- Apply prior knowledge of linear equations to construct and interpret lines of best fit to contextual data
- Use and evaluate inferences about the relationship between paired variables based on modelling data using linear relationships
- Compare mathematical linear relationships between variables to relationships between variables in the real world

ADVANCED PERFORMANCE CHARACTERISTICS

- Tolerate ambiguity and be able to work with inexact relationships
- Show insight into the nature of different types of linear relationships in mathematics and science

You will need This sheet, a clear ruler, a sharp pencil, and a spreadsheet or GeoGebra if allowed. All data are model values.

TASK 1 · PRACTICE · MATCH THE PAIRED VARIABLES

Independent *variables answer the question "What do I change?"* ***Dependent*** *variables answer the question "What do I observe?" Match the cards into nine pairs, with one card as the independent variable and one as the dependent variable. There is more than one way of pairing the cards.*

Temperature | Radioactive decay | Number of people in a lift | Life span | Mileage | The crop yield produced | Time in months | Seedling growth | Time in hours | Wind speed | Number of TVs | Total weight of passengers | Pressure | Children's heights | Time in years | Virus multiplication | Amount of fertiliser applied to a crop whilst growing | Petrol consumption

	INDEPENDENT VARIABLE (what I change)	DEPENDENT VARIABLE (what I observe)	THE CONTEXT I CHOSE AND WHY
1			
2			
3			
4			
5			
6			
7			
8			
9			

Compare with another pair. Which pairing was different from yours, and what context did they choose?

What if...? A card could go with more than one partner. Which card, and what are the two contexts?

Done when all nine pairs say which variable you change and which you observe, and you can explain a context for each.

TASK 2 · APPLY

Exact or Inexact?

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TASK 2 · APPLY · THREE GRAPHS, THREE KINDS OF RELATIONSHIP

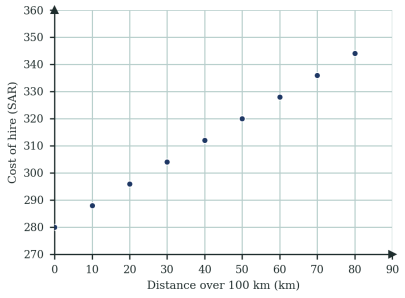
Consider the 3 graphs and accompanying explanations. Talk to your partner and together make comments about the relationship you see in each diagram and the context described. Try to use this vocabulary:

☐ exact linear relationship ☐ inexact linear relationship ☐ experimental errors ☐ limited assumptions

☐ complex assumptions

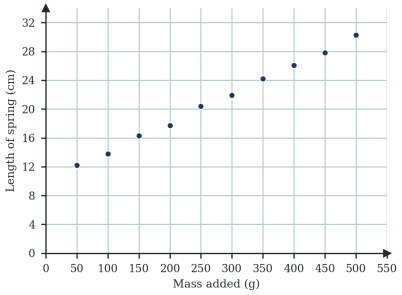
Graph A · Hiring a car in Jeddah

A car costs SAR 280 a day, the first 100 km is included, then SAR 0.80 for each extra km. The graph shows the cost for distances over 100 km.



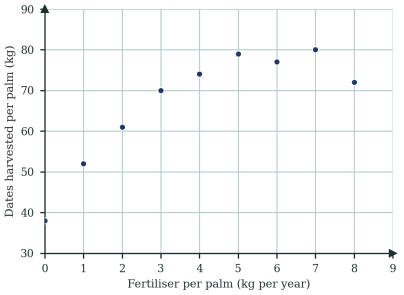
Graph B · Stretching a spring

Masses are added to a spring and its length is measured with a ruler each time.



Graph C · Fertiliser on date palms

A farm in Al-Ahsa records the fertiliser used on each palm and the dates harvested from it.



	Exact or inexact? Why?	Errors or assumptions	What the relationship tells me
A			
B			
C			

LOOK CLOSER AT GRAPH B

Use a ruler to draw a line of best fit. Choose two points ON your line (they need not be dots).

$(x_1,\ y_1) =$		$(x_2,\ y_2) =$	
$m = \dfrac{y_2 - y_1}{x_2 - x_1} =$		$c = y_1 - mx_1 =$	
my equation: $y =$			

Interpret. What does the gradient mean in the context of the spring? What does the intercept mean?

What if...? You hang 5 kg (5000 g) on the spring. What does your line predict, and should you trust it? Why?

Done when every graph is named exact or inexact with a reason, and you can interpret the gradient and intercept in context.

JUDGE IT, THEN APPLY

Test It and Use It

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JUDGE THE CLAIMS

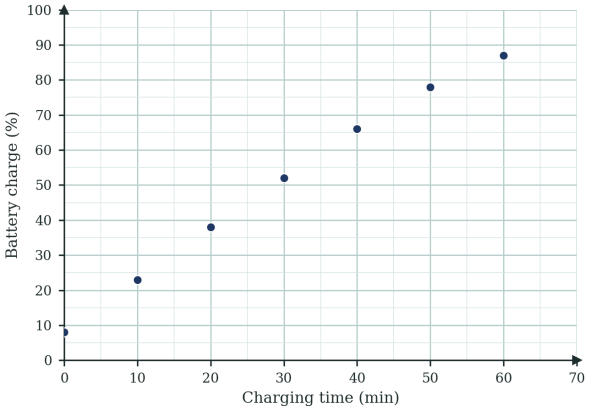
Decide whether each claim is true, false or sometimes true. Use evidence from your own work. One counterexample is enough to show a claim is false.

A	**If the dots do not lie exactly on a line, there is no linear relationship.**	☐ true ☐ false ☐ sometimes	*Because...* *Assumption I checked:* Does a model need to touch every dot to be useful?
B	**Only one way of pairing the cards is correct.**	☐ true ☐ false ☐ sometimes	*Because...* *Assumption I checked:* Did I say which context I chose?
C	**The line for the spring is reliable for any mass.**	☐ true ☐ false ☐ sometimes	*Because...* *Assumption I checked:* Does the spring still behave the same far beyond the readings?

Which graph gave you the strongest evidence? Was it exact, inexact, or one with complex assumptions? What made you decide?

APPLY AND PRESENT · A FAST CHARGER NEAR JEDDAH

Model values, not real charger data. An electric car is plugged into a fast charger and the battery charge is recorded every 10 minutes.



1 · Which variable is independent and which is dependent?

2 · Use the first five readings to draw a line and find its equation.

3 · Predict the charge after 25 minutes. Check on the graph.

4 · What if the car stays 90 minutes? Is the line still sensible? Say exact or inexact, and why.

My Mawhiba record. Tick the value I showed today: ☐ Inquiry ☐ Risk taking ☐ Perseverance ☐ Collaboration ☐ Concern for society
Example: .....